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New insights on traps states in organic semiconductor applying illumination-free transient current method

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ABSTRACT

A transient current measurement technique which does not need light stimulus is developed. This illumination-free transient current (IFTC) technique offers insights on trapping states characteristics such as the capture time of the trapping states, their density of states (DOS) and the attempt-to-escape frequency of trapped carriers. For the analysis of IFTC measurements carried out on metal–insulator–semiconductor (MIS) capacitors, numerical simulation of the transient current were performed as a way around usual assumptions of short transit times which can considerably distort the values from transient current measurements. A combination of experimental data and simulations revealed the need to take into account retrapping events in order to extract the correct band tail density of states and corresponding characteristic parameters. An exponential density of states with a total density of $4 \cdot 10^{23}$ m^{-3} and a width of 50 meV was found to be representative of the band tail of spin-coated poly(3-hexlythiophene) (P3HT). A broader exponential distribution with a width of 83 meV is extracted for the interface states towards SiO_2. Using a multiple trap and release model, the capturing time and the attempt-to-escape frequency of the band tail states were found to be 10^{-10} s and 10^{8} s^{-1}, respectively. With this technique, we are able to also pinpoint the energy position of the Fermi level relative to the transport energy level.
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1. Introduction

The first applications of organic field effect transistors (OFETs) in e-paper displays, simple integrated circuits, and chemical and biological sensors have been recently demonstrated [1–4]. Even with OFETs making their path to real applications, understanding charge transport in these disordered materials is still not fully achieved.

A crucial obstacle to the full understanding of charge transport in organic semiconductor based devices is the characterization of the band tail states responsible of the charge transport in organic semiconductor devices. In fact, the main obstacle to high mobilities in organic semiconductors is the slowing down of charge carriers due to hopping and/or trapping of charge carriers in the band tail states [5]. These trapping states are not only characterized by their density but also by their capture and emission times. Therefore a better future for organic electronics requires tools to quantify all parameters relevant for charge transport properties in all environments. In order to obtain information on the trapping process in organic semiconductor based thin films several methods have recently attracted attention or are commonly in use. Deep level transient spectroscopy (DLTS) [6], thermally stimulated current (TSC) [7], photo-excited charge collection spectroscopy [8], impedance spectroscopy (IS) [9,10], transient photocurrent spectroscopy (TPC) [11] and time-of-flight (TOF) method [12] have been used with partially satisfactory results. The first three methods are rather complex and often inadequate for organic semiconductors due to the involvement of bias and temperature stress for DLTS and TSC and light stress for photo-excited charge collection spectroscopy. Though, the latter two methods offer several advantages from the simplicity of the measurements to the wide range of accessible information on the semiconductor properties, they still have their limitations. For IS, the need of complex equivalent circuits for the analysis restrains the accessible information for the band tail characterization [13]. TPC and TOF use a structure with a semiconductor sandwiched between two metal contacts and additionally requires light illumination, a parameter known to be a source of instability due to its potential doping effect on some organic semiconductors [14]. For TOF measurements the layer thickness must be much larger than the exciting light penetration depth, which is often much thicker than in device structures. In this paper, TPC and TOF methods were modified so that light illumination is avoided. The device of choice is a MIS capacitor. The use of a MIS capacitor opens new doors with a modified illumination-free transient current (IFTC) measurement technique.

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In this new transient current measurement approach, the charge carriers packet to be transported is created at the semiconductor–insulator interface and transported through the semiconductor layer upon applying a bias step function. The transport of charge carriers through the semiconductor layer can be analyzed and subsequently reveals information on the physical properties of the interfaces and the bulk of the semiconductor.

In this paper, we provide a full description of transient current measurements carried out on organic semiconductor based MIS capacitors. The analysis shines light on crucial physical properties of the band tail states such as the density of states, the capture time associated to these trapping states and the attempt to jump frequency of a trapped carrier. Section 2describes the experimental details. In Section 3we provide a numerical model for transient current measurements simulation. In Section 4we present the experimental results according to the theoretical background developed in Section 3.

2. Experiment

A highly n-doped silicon (Si) wafer (resistivity = 2 mΩcm) with a 100 nm thick thermally grown silicon oxide (SiO_2) was used as substrate for the MIS capacitor (Fig. 1). The highly n-doped Si serves as gate and SiO_2 as insulating gate dielectric. The substrate was first wet chemically cleaned in an ultrasonic bath with solvents (acetone, isopropylacetate) and treated in an ozone cleaner for 10 min. A 170 nm thick poly(3-hexylthiophene) (P3HT) (Sepiolid-P100, Rieke Metals, regioregularity ~95%, average molecular weight $M_w = 5 \times 10^4$ g/mol) layer was then spin coated onto the cleaned substrate from a 1.5 wt.% chloroform solution. The top contact of the MIS capacitor was made by sputtering a 100 nm thick top ohmic gold (Au) contact onto the semiconductor.

The transient current measurements setup is shown in Fig. 1. As the setup reveals, a step bias function is applied on the gate terminal of the MIS capacitor, subsequently creating a charge carrier packet. The created packet of carriers moves across the semiconductor layer inducing a time dependent current. This current is measured through the load resistance (R_{Load}) with a 100 MHz bandwidth digital oscilloscope. As the measured transient current decreases with time, in order to extend the measurements time range and still be able to measure low currents, the load resistance was varied between 10 Ω and 100 kΩ. A time span of 10 ns to 100 ms could be obtained. The capacitive effect of the device produces a short RC time constant that was included in the measurements analysis. Depending on the load resistance, the RC time constant ranges from 10 ns to 100 μs. The step function was generated by a Keithley 3390 function generator. The effectively applied voltage to the MIS structure was verified using the second channel of the oscilloscope (not shown in Fig. 1). 64 discrete events were averaged to increase the 12-bit resolution of the oscilloscope. The time interval between two successive pulses was set to 10 s and for each series of pulses the background current was recorded and then subtracted from the obtained transient current.

Temperature dependent measurements were carried out in a cryostat with the sample mounted in vacuum on a cold-finger and cooled down with a closed loop liquid Helium supply.

3. Theoretical description

3.1. Time of flight (TOF) and transient photocurrent (TPC) measurements

Our approach is a modification of the usually used time of flight (TOF) and transient photocurrent (TPC) techniques. TPC and TOF measurements are widely used to measure the mobility μ and the density of band gap states (DOS) in amorphous inorganic and organic semiconductors [11,15]. In TOF measurements, a thin sheet of charge carriers is photogenerated close to the contact, while one

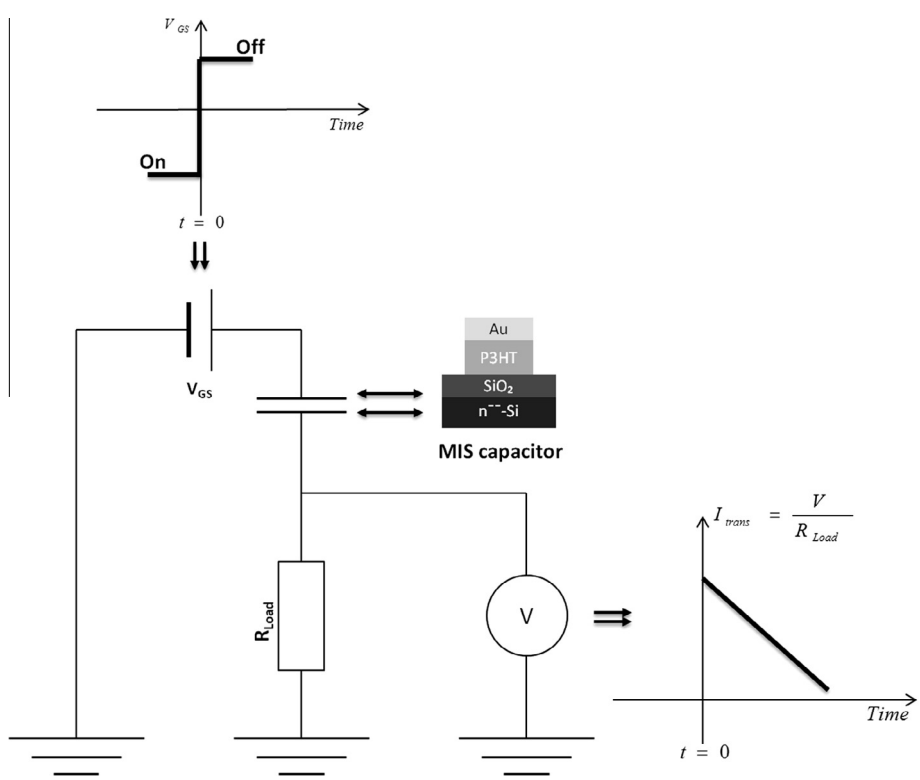

Fig. 1. The setup used for transient current measurements is shown together with a step function applied on the MIS capacitor and the current measured across the load resistance.

type of charge carrier is rapidly collected by the closest contact, the other type is transported across the sample and hence dominate the transient current. In TPC measurements, charge carriers are photogenerated across the whole sample and both type of charge carriers will be collected. However, the transient current will be dominated by the slower type [12].

Despite the differences between the two techniques, the core of the resulting transient current is similar. Whether it is TOF or TPC, while the charge carriers propagate through the device, some carriers are trapped. The trapped carriers are released later than the transit time of the fastest carriers, generating a delayed current dominated by charge carriers detrapping. A typical current trace as it would be measured in TOF or TPC measurements is schematized in Fig. 2. The transient trace can be divided in three regions [15,12,11]:

- The pre-transit region (region I): In TOF, this region can be understood as the short time range in which carries under investigation have not left the sample. The induced current corresponds to the drift of the charge carriers packet towards the collecting electrode. This current decreases with time due to the immobilization of carriers by trapping.
- The transit time (region II): The transit time (t_T) corresponds to the time at which the leading edge of the charge carriers packet reaches the collecting contact and thereafter a major drop of the current level occurs.
- The post-transit time (region III.a & III.b): The post-transit current is the time region where the current is dominated by the release of charge carriers from traps. At the beginning, it consists of the release from shallow traps (region III.a). While it is later dominated by the release from deep traps (region III.b).

With an appropriate transformation of the current in region III, the DOS as a function of energy can be estimated from both types of measurements [15–17]. For TOF measurements, the mobility of the fastest carriers can be additionally estimated from the transit time. The commonly used method to estimate the DOS from the obtained transient current trace is the one proposed by Seynhaeve [15]. In this method, a given measurement time t in region III corresponds a demarcation energy $E_D(t)$ separating states that are located above $E_D(t)$ and that have a very high probability of being already released of trapped carriers at the time t, from the states below $E_D(t)$ where trapped carriers are very unlikely to be released before the time t,

$$E_D(t) = kT\ \ln(v_o t), \tag{1}$$

where T is the absolute temperature, k is the Boltzmann's constant and v_o is the attempt-to-escape frequency. The transient current at a given time t is directly proportional to the product of the charge carriers at the demarcation energy ($ed_sAfN(E_D)$) and the rate at which the demarcation energy changes (dE_D/dt),

$$I(t) = ed_sAfN(E_D)\frac{dE_D}{dt}, \tag{2}$$

where e is the elementary charge, d_s is the thickness of the semiconductor, A is the device area, $N(E_D)$ is the density of states at the demarcation energy and f is the fraction of occupied states and is assumed to be energy independent. Thus, the DOS is related to the post-transit current via:

$$N(E) = \frac{tI(t)}{ed_sAfkT}. \tag{3}$$

Though the fraction of occupied states f is difficult to estimate, it is usually set to one as long as it is experimentally assured that all trap states are filled. The latter fact can be verified by observing experimentally the saturation in the transient current upon increasing the generated free charge carriers [11]. Therefore, the calculated DOS becomes

$$N(E) = \frac{tI(t)}{ed_sAkT}. \tag{4}$$

For the specific case of exponential DOS, commonly assumed to be representative of the band tail states in organic semiconductors [18],

$$N(E) = \frac{N_t}{E_o}\exp\left(-\frac{E}{E_o}\right), \tag{5}$$

the transient current follows the already known power law characteristic of a dispersive transport [15,19,20]

$$I(t) = ed_sAf\alpha N_t v_o^{-\alpha} t^{-\alpha-1}, \tag{6}$$

where E_o is the steepness of the rate at which the traps density decline with increasing energy, N_t is the total number of band tail states per unit volume and α is a dispersive parameter which can be correlated to E_o by

$$\alpha = \frac{kT}{E_o}. \tag{7}$$

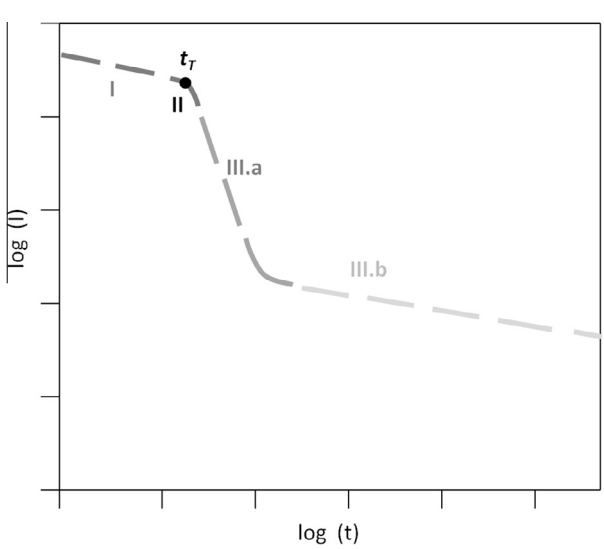

Fig. 2. Schematic of typical transient current trace.

3.2. Illumination-free transient current measurements (IFTC)

Our approach follows the general approach of purely electrical measurements. An electrical stimulus is applied to the electrodes and the current response of the device is subsequently observed. In our case, we apply a voltage step function

$$V(t) = \begin{cases} V_{on} & \text{for } t < 0, \\ V_{off} & \text{for } t \geqslant 0 \end{cases} \tag{8}$$

to the device and we measure the resulting transient current $I(t)$ for $t \geqslant 0$. V_{on} usually accumulates charges in the semiconductor at the insulator-semiconductor interface. The step between the two bias voltages creates a charge carrier packet which subsequently moves through the device. The resulting transient current is a fingerprint of the device intrinsic properties.

The conditions in our IFTC approach can alter the measurements outcome usually seen in TOF and TPC outcome:

- While a sample sandwiched between two coplanar electrodes is the device of choice for TOF measurements and TPC, a MIS capacitor is used for this approach.
- While the charge carriers packet is photogenerated for TOF and TPC measurement, it is generated by a voltage step in our approach.
- While the charge carriers packet drifts under a mainly homogeneous and time independent field for TOF and TPC measurements, the field is more inhomogeneous and time dependent in this new approach.
- For a MIS capacitor, the applied voltage will drop mainly at the insulator.
- Spurious carrier injection and leakage currents are avoided by the extra insulator compared to potentially not optimal blocking contact properties in TOF and TPC. Therefore deep traps signatures corresponding to smaller currents at longer timescales are easier to access.

In order to evaluate the applicability of the method and subsequently extract the relevant parameters, numerical simulations of the device response were done. For the numerical simulations, we used a 1-d drift–diffusion device model, where the device was modeled using a discrete spatial grid (as illustrated in Fig. 3(b)). At each point of the grid, a set of spatial properties are assigned, doping concentration $N_A(x)$, mobility $\mu(x)$ and trap density $g(E,x)$.

The charge carrier packet to be transported through the sample can be visualized as the difference between the charge carrier distribution before and after the bias step. The excess charge carrier distribution generated at the semiconductor–insulator interface in accumulation by a voltage V applied across a MIS capacitor was calculated by Mott and Gurney in 1940 [21,22],

$$p(x) = \frac{p_o}{\left(1 + x/\sqrt{2}L_D\right)^2}, \tag{9}$$

with the Debye length

$$L_D = \frac{\sqrt{2}kT\epsilon_s\epsilon_o}{eC_i(V - V_{FB})} \tag{10}$$

and the carrier concentration at the interface

$$p_o = \frac{eC_i^2(V - V_{FB})^2}{2kT\epsilon_s\epsilon_o}, \tag{11}$$

where ϵ_o is the vacuum permittivity, ϵ_s is the semiconductor dielectric constant, C_i is the insulator capacitance and V_{FB} is the flat-band voltage. The initial total charge carrier density is essentially the sum of the charge carrier generated by the applied voltage $p(x)$ and the fixed background of free charge carriers from the doping $N_A(x)$ minus the total trap states density.

The effect of trapping is taken into account by including a probability to capture and to release free carriers to and from trap levels i, each having a density of states $g_i(x)\Delta E$ and an energy $E_i(x)$. Each trap interacts directly and locally with the transport energy (assumed to be the zero energy). The dynamic of the trapping process is given by the capture rate of traps parametrized by a common capture time constant τ_c. τ_c is defined for a given reference free charge carrier density p_{ref} as only the product $\tau_c p_{ref}$ has a practical meaning. In this paper p_{ref} was set to be equal to the doping concentration N_A for standard reasons [23]. The capture rate is given by,

$$a(x) = p(x)\sum \frac{1}{p_{ref}\tau_c} g_i(x)\Delta E(1 - f_i(x)) \tag{12}$$

and the release rate of the traps parametrized by a common attempt-to-escape frequency v_o,

$$u(x) = \sum g_i(x)\Delta E f_i(x) v_o \exp\left(-\frac{E_i(x)}{kT}\right), \tag{13}$$

where $f_i(x)$ is the occupation number of trap level i and has values between 0 and 1.

The electric field distribution $\vec{F}$ is obtained from the electrical potential,

$$\vec{F} = -\nabla\phi \tag{14}$$

and the electrical potential ϕ is calculated via the Poisson equation from the charge density $\rho(x)$

$$-\nabla^2\phi = \frac{\rho(x)}{\epsilon_s\epsilon_o}. \tag{15}$$

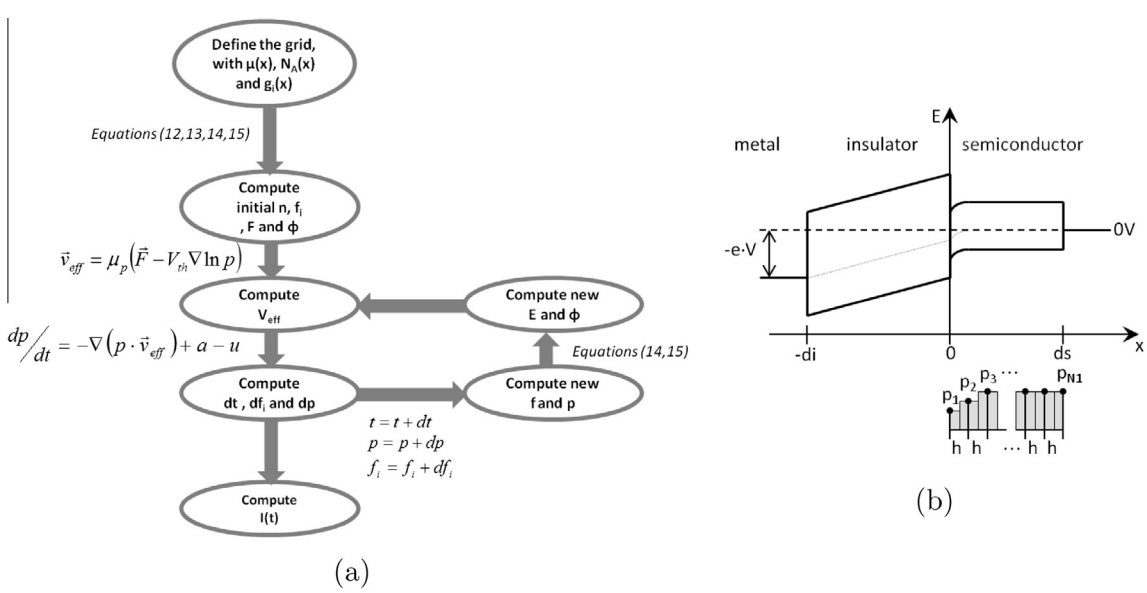

Fig. 3. (a) Diagram of the transient current simulation flow. Depending on the complexity assumed, the simulation requires computational times of minutes up to 5 h on a standard pc using a single CPU core. (b) The MIS capacitor as it was divided in a discrete spatial grid with stepsize h within the semiconductor for the numerical simulation.

The electric field at $x = 0$ is specific for a given charge distribution and the outside voltage drop over the MIS capacitor. The set outside voltage represents the boundary conditions for the solution.

The time evolution of the system is described by the drift and diffusion of the charge carrier packet including trapping and detrapping dynamics through the continuity equation,

$$\frac{\partial p}{\partial t} = -\nabla(\vec{j}/e) + a - u, \tag{16}$$

where the current density ($\vec{j}$) is given by the drift and the diffusion of not trapped charge carriers,

$$\vec{j} = ep\mu_p F - e\mu_p V_{th}\frac{\partial p}{\partial x}, \tag{17}$$

with μ_p and $V_{th} = kT/e$ being the mobility and the thermal voltage, respectively. Effects of minority carriers are neglected. The simulation flow of the transient current of the majority carriers (holes) is sketched in Fig. 3(a).

4. Results and discussion of experimental results

4.1. Interfacial states density extraction

Eq. (4) is commonly used to extract the density of the trapping states though it neglects retrapping and the extraction time. In this subsection, we use simulations to evaluate in which extent the density of the trapping states at the insulator-semiconductor interface can be extracted following this simplified equation. We consider a sample with fixed geometrical parameters, across which we apply a voltage step function. The assumed bulk conductivity is set in the range of typical values for P3HT ($10^{-3} - 10^{-6}$ S/m) [24,13]. The overall input parameters used for the simulations are given in Tables I, II and III. For simulations of energetically continuous density of states, the DOS was represented by a closely spaced set of m discrete levels with a spacing of $\Delta E = eV_{th}$ and a density $g_i = g(E_i)\Delta E$. In order to understand the trapping and release process of carriers, two cases were considered. Firstly, a distribution of traps located only at the interface was considered. Secondly, a single level trap uniformly distributed in the bulk was then added to the first case. The first case was used as a simple proof of the applicability of Eq. (4) for interface states density extraction. The second case evaluate the effect of the retrapping in band tail states of charge carriers released from interface trapping states on the delayed current from interface states.

For the first case, we looked at the possibility of retrieving the assumed density of states using Eq. (4). The transient current resulting from the release of trapped carriers corresponds to the tail seen after $t = 5\ \mu s$ as illustrated in Fig. 4(a). A DOS was extracted from the obtained transient current via Eq. (4) and compared to the input DOS in Fig. 4(b). For deep traps, we find that the input and the calculated DOS overlap. For more shallow traps, with release times becoming comparable to the transit time, deviations can be seen. The latter region corresponds to the transition between the charge carriers drift limited current and the charge carriers detrapping limited current. Therefore, for deep traps, with a release time clearly larger than the transit time, the interface states DOS can be estimated from the measured transient current. This simulation shows that the typical transformation via Eq. (4) from a current transient dominated by the release from deep traps to a DOS is still sufficient in order to estimate the interface states distribution, despite the inhomogeneous and time-dependent field distribution occurring in IFTC. Nevertheless, it is also crucial that we analyze the effect of the presence of band tails in the bulk as the transient current could be distorted as a result of trapping and retrapping events in the bulk of the carriers released from interfacial traps. The transient current was computed for the second case with an additional single shallow trap level at $E_t = 0.05$ eV uniformly distributed in the bulk with a density of $N_t = 10^{23}$ m^{-3}. The used density of traps is a typical density of traps in organic semiconductor [11]. Fig. 4(c) shows that though the transient current is slightly affected, the tail due to the release of trapped carriers from deep interfacial traps is unperturbed. The only seen effect is the shift of the beginning of the tail. The shift of this point to longer times depends on how deep are the bulk traps and their density. The deeper the bulk trap states, the smaller becomes the available energy range for interface states analysis. Hence, the experimentally obtained transient current can be analyzed according to Eq. (4) as long as the transit time and the trapping time of the bulk band tail states remains small compared to the detrapping time of the interfacial states.

4.2. Transient response in accumulation or depletion regime

Depending on whether or not the final voltage keeps the device in accumulation or drives the device in depletion, transient current measurements on MIS capacitors access different information. Impedance measurements showed that the AC response will include information on band tail states responsible of the dispersion of charge transport mobilities (dispersive transport regime) if the device is biased in accumulation. The AC response will contain additional information on the deep trap states if the device is biased in depletion [9,13]. Fig. 5 shows the same behavior for our experimental transient current curves. If the final applied bias keeps the device in accumulation (Fig. 5(a)), the transient current gives only a picture on the charge carrier transport in the band tail states through the bulk and this is translated by the early current switch off at around $t = 1$ ms as the transient current switches off once the demarcation energy reaches the Fermi energy. If the final applied bias drives the device into depletion (Fig. 5(c)), the transient current is additionally affected by the release of carriers from even deeper trapping states in the depletion zone and this is translated by the missing switch off of the transient current at very long time scales. The appearance of this tail in the transient current can be used as a tool to investigate interfacial trap states.

Table I
General geometrical and dielectric parameters of MIS device and simulation stepsize.

A (mm^2)	d_i (nm)	ϵ_i	d_s (nm)	ϵ_s	h (nm)
10	100	3.9	170	2.8	0.8

Table III
Step function shape for testing Eq. (4)

V_{FB} (V)	V_{on} (V)	V_{off} (V)
0	−10	+10

Table II
Simulation parameters assumed for testing Eq. (4).

N_t (10^{23} m^{-3})	E_o (meV)	v_o (10^8 s^{-1})	τ_c (10^{-8} s)	N_A (10^{24} m^{-3})	μ (10^{-9} m^2/V s)
1	70	1	1	1	1

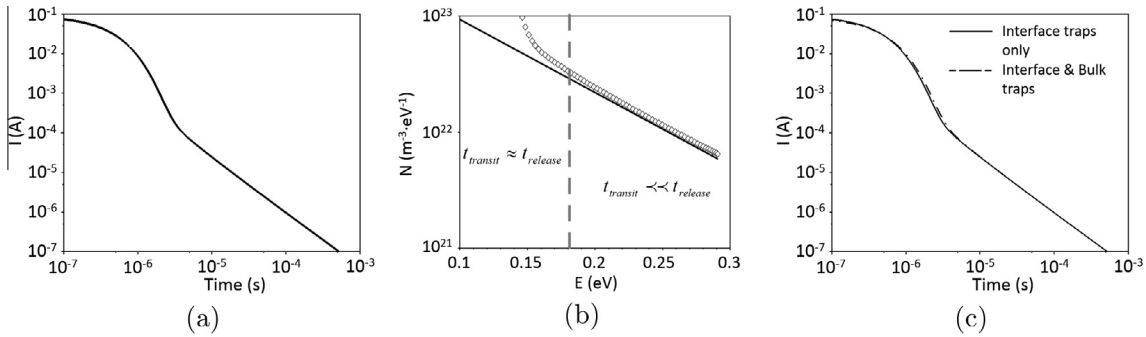

Fig. 4. (a) Numerically computed transient current using the parameters in Tables I–III. (b) Comparison of the input DOS (line) and the output DOS obtained using Eq. (4) (symbol) on the transient current. (c) Distortion of the transient current represented in (a) upon adding a bulk trap level located at $E_t = 0.05$ eV and with $N_t = 10^{23}$ m^{-3} to the system represented in (a).

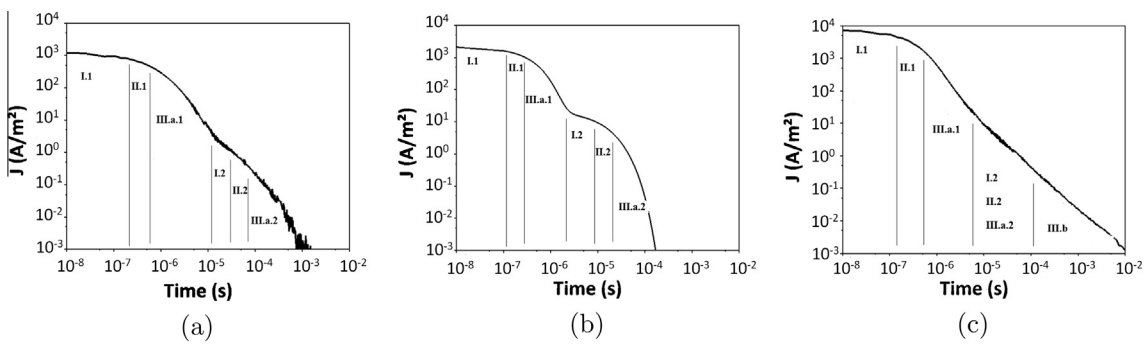

Fig. 5. Final bias regime dependence of the transient current. (a) Measured transient current with the bias step function keeping the device in accumulation ($V_{on} = -10$ V and $V_{off} = -5$ V). (b) Computed transient current if a step function in accumulation ($V_{on} = -10$ V and $V_{off} = -5$ V) is a applied to a sample with two layers of different conductivities ($\mu_1 = 10^{-9}$ m^2/Vs, $\mu_2 = 10^{-11}$ m^2/Vs), ($d_1 = 155$ nm and $d_2 = 15$ nm). (c) Measured transient current if the bias step function drives the device into depletion ($V_{on} = -10$ V and $V_{off} = +10$ V).

In TOF and TPC measurements, the applied constant voltage does not allow the switching off of the transient current once the demarcation energy reaches the Fermi energy. In ITFC, we are able to probe the energy difference of the transport level and the Fermi energy by looking at the switching off point of the transient current in accumulation. Transient currents were measured for different doping concentration. In order to dope the semiconductor, the semiconductor was exposed to light and oxygen. As shown in Fig. 6(a), an increase of the doping concentration leads to an earlier switch off of the transient current as the Fermi energy is shifted closer to the transport energy. The energy difference between the Fermi level E_F and transport level E_T was calculated from the switch off time using Eq. (1). For an exponential DOS, the doping concentration should be proportional to exponential of the energy difference $E_F - E_T$ and this is what is seen in Fig. 6(b) for low doping. For the last point at high doping concentration, a deviation is seen and this might be due to the accuracy in pinpointing the switch off time.

The same main regions as seen in a typical transient current for TOF and TPC measurements (Fig. 2) can also been seen in both cases. However, dividing the trace current in the three regions requires more attention as one has to consider one crucial effect which could potentially alter the obtained transient current. As shown by previously published impedance measurements, during

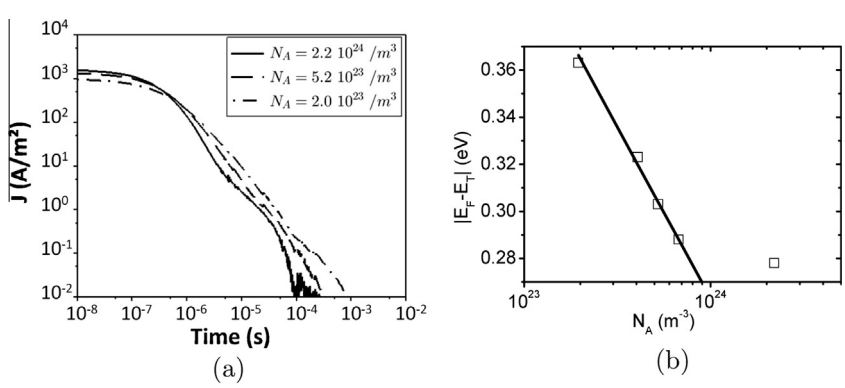

Fig. 6. (a) Doping concentration dependence of the transient current. The doping concentration N_A was calculated from the capacitance–voltage measurements. (b) Exponential relation between $E_F - E_T$ and the doping concentration. The squares correspond to the experimental values and the solid line corresponds to an exponential relation.

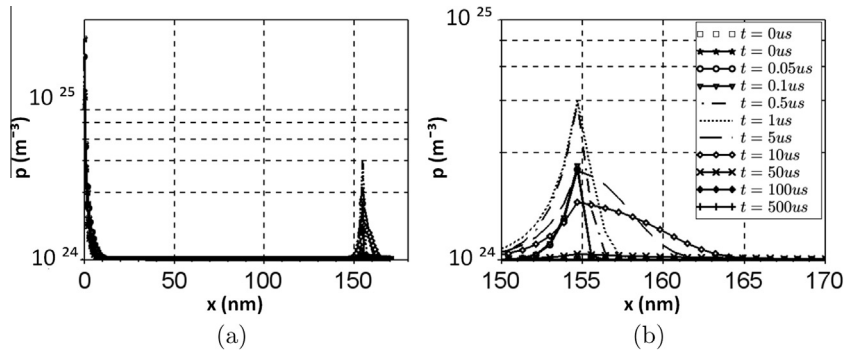

Fig. 7. Dynamic of the charge carrier packet through the semiconductor layer for the transient current simulation represented in Fig. 5(b). In (a) we show the charge carrier packet in the whole semiconductor layer at different times and in (b) we zoom in at the interface of the two layers.

sputtering of the top contact, Au particles penetrates the semiconductor below and this creates a small less conductive layer close to the contact [13]. In fact, the transit time does not correspond to the first drop of the current but rather to the second drop. The existence of the two slopes in the pre-transit region is a direct consequence of the slowing down of charge carriers in the less conductivity region observed close to back contact [13]. Fig. 5(b) shows the simulated transient current of a system with two layers of different conductivity, a layer with the normal conductivity ($\mu = 10^{-9}$ m^2/V s) and a layer with a reduced conductivity ($\mu = 10^{-11}$ m^2/V s) of 15 nm thickness located at the top contact. The simulated transient current shows the same behavior as the experimentally obtained transient current in the accumulation case. Assigning the two main current drops to the transit times in the two layers can be done via following the dynamic of the charge carrier packet through the sample. Fig. 7 shows the dynamic of the charge carrier packet in the less conductive layer. It shows that region I.1 in Fig. 5(b) corresponds to the movement of the packet in the first layer as the density of charge carriers accumulated at the interface of the two layers stays constant between $t = 10^{-8}$ s and $t = 10^{-7}$ s (see Fig. 7(b)). Region II.1 corresponding to the first drastic decrease of the current corresponds to the arrival of the charge carrier packet at the interface between the two layers and this is translated by the sudden increase of charge carrier density at the interface of the two layers observed between $t = 10^{-7}$ s and $t = 10^{-6}$ s (see Fig. 7(b)). Region I.2 corresponds to the movement of charge carrier packet in the less conductive layer and this is seen by the decrease of the charge carrier density at the interface between the two layers. The last major drop of the current (II.2) takes place once the charge carriers reaches the collecting electrode. Regions III.a.1 and III.a.2 are dominated by the release of charge carriers from the shallow traps in the first and second layer, respectively. Region III.b corresponds to the release of carriers from deep traps and is only seen in depletion. This region III.b is used later for deep traps analysis.

4.3. Field dependence of the transient current

A crucial assumption in the commonly used analysis of TOF and TPC measurements via Eq. (4) is that the electric field in the semiconductor is high enough to assure that the release time of delayed carriers is extremely long relative to the transit time of carriers. Furthermore, it is assumed that all trapping states are fully occupied after the transit of the fast carriers. While these assumptions are pretty reasonable for deep traps with long release times, they are not easy to guarantee for carriers from shallow band tail states giving the main signature of charge transport in disordered materials [15]. The uncertainties originating from these assumptions cast a doubt on the use of Eq. (4) in order to estimate the band tail states DOS from the post-transit current transient for more shallow traps.

A look at the field dependence can be used as a quick way to experimentally investigate in which limits our transient current measurements are affected by the transit time. The transit time can be experimentally changed for a fixed voltage step via varying the thickness of the semiconductor. In Fig. 8, the transient current dependence on the active layer thickness was investigated. If the final voltage of the step function keeps the device in accumulation as done in Fig. 8(a), the obtained transient current is clearly affected by increased layer thickness. The initial current decreases due to the higher resistance of the thicker layer. A clear shift to longer times of the transit time in the first layer is evident. The transit in the second layer undergoes negligible changes as the thickness of this second layer determined by the electrode

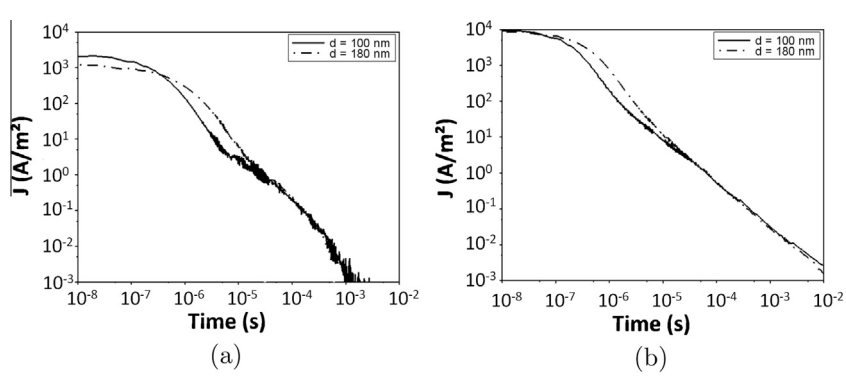

Fig. 8. (a) Thickness dependence of the transient current if the bias step function keeps the device in accumulation ($Von = -10$ V and $V_{off} = -5$ V). (b) Thickness dependence of the transient current if the bias step function drives the device from accumulation into depletion ($V_{on} = -10$ V and $V_{off} = +10$ V).

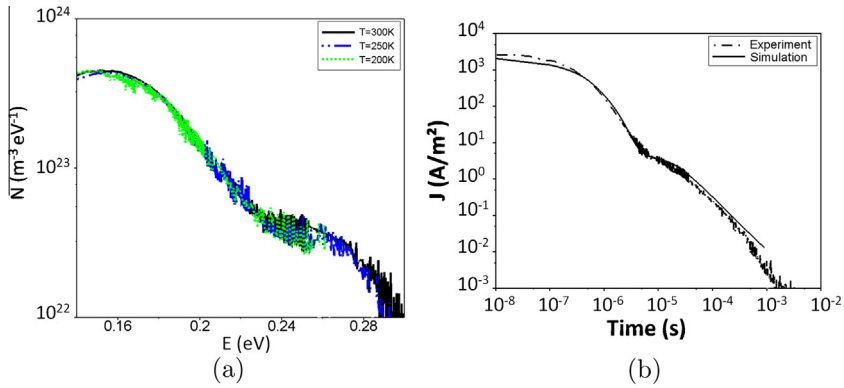

Fig. 9. (a) DOS plotted from transient currents measured at different temperatures (transformed via Eqs. (1) and (4)) using $\nu_o = 10^8\ s^{-1}$. (b) Experimental data closely matched with simulation.

Table IV
Parameters used to reproduce the experimental data as illustrated in Fig. 9(b).

N_t (10^{23} m^{-3})	E_o (meV)	ν_o (10^8 s^{-1})	τ_c (10^{-8} s)	N_A (10^{24} m^{-3})	μ_1/μ_2 (10^{-9} m^2/V s)
4	50	1	0.01	0.7	1.5/0.016

deposition process stays the same. If the final voltage of the step function drives the device in depletion as done in Fig. 8(b), a similar outcome as before, i.e. a finite current at long time scales, is obtained. A field dependence of the transit in the first layer is again observed. However the long tail corresponding to release of trapped carriers from deep traps is clearly independent on the field as one would expect. Hence it is clear that the release of trapped carriers in deep traps can be analyzed according to Eq. (4). Nevertheless, analyzing the shallow band tail states will requires a full simulation as the short release time transient current resulting from the shallow band tail states is clearly affected by the increase of the transit time.

4.4. Simulation: band tail states analysis

Full simulation of the system is a powerful alternative to Eq. (4) for the analysis of the shallow band tail states. We apply the numerical simulation to determine a band tail DOS that leads to a simulated trace current closely matching the measured current trace. For the simulation, the main input parameters are the doping concentration, the mobility in the bulk and in the region close to the top contact, the capturing time, the attempt-to-jump frequency and the DOS of the band tail states. The doping concentration is extracted from C(V) measurements carried out on the MIS capacitor. The ratio between the two different mobilities in the two layers is also known from Z(f) measurements. In order to estimate the attempt-to-escape frequency ν_o required to calibrate the energy positions, transient current measurements were carried out at three different temperatures and ν_o was adjusted in a such away that the effective density of states given by Eq. (4) is the same for the three considered temperatures (see Fig. 9(a)). With ν_o we adjust the energetic offset of the curves in Fig. 9(a) via Eq. (1). A value of $10^8\ s^{-1}$ was obtained, which is comparable to other organic materials as e.g. $10^7\ s^{-1}$ for a PDPP2FT/PC71BM blend [12]. It is not surprising that the obtained value for P3HT is lower compared to the $10^{12}\ s^{-1}$ reported for amorphous silicon based devices as P3HT has a lower conductivity [15]. The parameters to be provided for the simulation analysis are the capturing time and the DOS of the band tail. An exponential DOS was assumed as it has been proven that an exponential DOS is representative of the band tail in disordered systems [18]. As it can be seen in Fig. 9(b), a good agreement between the experimental data and the simulation can be obtained. The good agreement was reached by assuming a capturing time of 10^{-10} s for a p_{ref} of $7 \cdot 10^{23}\ m^{-3}$ and a exponential band tail with a number of states of $N_t = 4 \cdot 10^{23}\ m^{-3}$ and a width parameter of $E_o = 50$ meV (see Table IV). The obtained values confirm the reported values of E_o between 40 and 60 meV and N_t between 10^{23}–$10^{24}\ m^{-3}$ for the band tail of P3HT [18,25,26].

4.5. Interfacial deep trap states analysis

As already shown, if the device is driven into depletion, the transient current at long transient time t $\geqslant 10^{-5}$ s is dominated

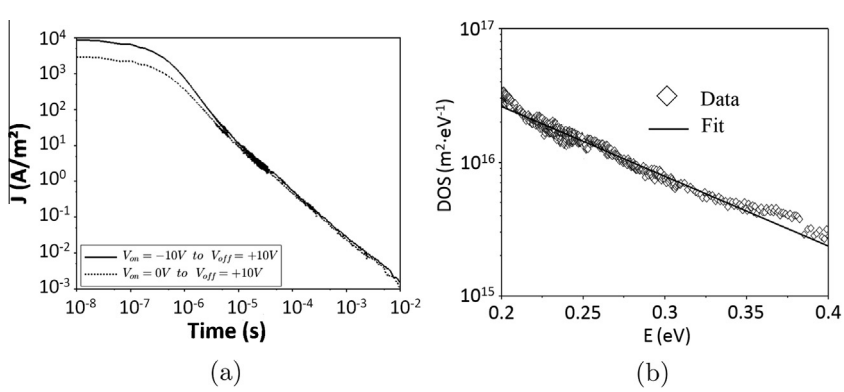

Fig. 10. (a) Step height dependence of the transient current. (b) The interfacial DOS calculated from the transient current in (a) using Eq. (4) (diamonds) is fitted with an exponential DOS (solid line).

by the release of carriers from deep interfacial trapping states and can be extracted via Eq. (4). The interfacial trapping states can be considered completely filled as the long tail resulting from emptying these deep states does not depend on the bias step function height in the considered range (a fact illustrated in Fig. 10(a)). The density of states obtained after transforming the current with Eq. (4) is plotted in Fig. 10(b). The energy scale corresponds to the results of the temperature dependent measurements as previously shown. For times beyond 10^{-5} s, corresponding to $E = 0.225$ eV, the DOS was fitted by an exponential decay and results in a characteristic steepness of 83 meV and a total density of states of $2.4 \cdot 10^{16}\ m^{-2}$.

5. Conclusion

In this paper, we present a light stimulus free transient current measurements technique. This technique is successfully carried out on a polymer based MIS capacitor. With this technique we were not only able to access parameters such as the DOS shape but also not easily accessible parameters such as the capturing time and the attempt-to-jump frequency. Furthermore, we show that this technique is able to provide insights on the properties of the shallow band tail states in the bulk, on the properties of the interface states at the semiconductor–insulator interface and on the Fermi level position relative to the transport level. Various simulations combined with experiments are utilized to assure the validity of this technique as a credible tool for DOS parameter extraction.

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